

Multiple Choice (10 marks)

1. Connor is measuring the capacity of a jar. Which unit of measure can he use?
 - (a) Centimeter
 - (b) Cup
 - (c) Gram
 - (d) Pound
2. Evaluate $-7(y - 6)$ for $y = 8$.
 - (a) -14
 - (b) 50
 - (c) -13
 - (d) -62
3. Let $a \star b = a^b - ab$. If $2 \star x = 22$, find x .
 - (a) 5
 - (b) 22
 - (c) 11
 - (d) 6
4. What is the minimum value of $a^2 + 6a - 7$?
 - (a) -30
 - (b) -12
 - (c) -7
 - (d) -16
5. Cat food costs \$.47/lb. How much does a 6-lb bag cost?
 - (a) \$2.82
 - (b) \$2.97
 - (c) \$6.47
 - (d) \$12.77

True / False (10 marks)

Answer True or False

6. True or False: The area of the triangle with altitudes of lengths 10, 12, and 15 is $2407/7$ square units.
7. True or False: A convex hexagon can have 6 distinct diagonals drawn from its vertices.
8. True or False: The power of the test, which is the probability of correctly rejecting H_0 when H_0 is false, is 0.74.
9. True or False: The probability of drawing the balls in alternating colors from a box of 4 white and 4 black balls is $\frac{1}{2}$ of the total arrangements.
10. True or False: The solution to the equation $-2k = -34.8$ is $k = -17.4$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given that the mean daily demand for bread at a bakery is 2,500 loaves with a standard deviation of 225 loaves, analyze the likelihood of the bakery running out of bread when it bakes 3,000 loaves. Discuss your calculations and reasoning.
12. Calculate the total number of chairs set up for a meeting if there are 6 rows with 8 chairs in each row. Explain your reasoning and the steps you took to arrive at your answer.

End of Paper